

# Biological Foundations and Regulation

Genome, epigenome, cells, metabolism, signaling, tissues, microbiomes, development, aging, and systems biology

Volume 02 of the Health Atlas Research Suite

## Purpose

Provide the shared mechanistic substrate beneath organ systems while distinguishing causal biology from high-dimensional correlation and molecular marketing.

**Evidence, guidance, and standards cutoff: 13 August 2026**

August 2026

# Document status and intended use

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This is a technical evidence synthesis and research-engineering reference, not individualized medical advice, a substitute for clinical evaluation, or authorization to use an experimental device on people. It distinguishes established knowledge from uncertain inference and frontier research. Diagnostic and therapeutic decisions require qualified professionals, appropriate reference standards, and applicable ethics and regulatory review.

## Suite relationship

This volume is read with the Atlas of Human Health (master taxonomy), the Library Guide and Crosswalk, and the other domain volumes identified in the cross-reference chapter. Volume 03A, Cardiovascular Health and Measurement Engineering, remains the reference model for the suite.

## Evidence boundary

This volume is a mechanistic research framework, not a guide for ordering consumer genomic/omic tests or using unapproved molecular therapies. Molecular association does not establish diagnosis, cause or treatment benefit.

# How to use this volume

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Human biology is a nested adaptive system: sequence and regulation shape cells; cells form tissues; tissues exchange matter, force and signals; organs and behavior operate within social and physical environments. No molecular layer is inherently more “real” than function, and multi-omics does not create causality by volume. Useful translation requires an explicit chain from mechanism to measurable state to decision and patient-important outcome.

## Questions this volume is designed to answer

1. How do levels from genome to whole person constrain and influence one another?
2. Which molecular findings are causal, prognostic, predictive or merely correlated?
3. What can current omics and models reliably measure?
4. How should cell/gene/regenerative technologies be validated?
5. Where does reductionism fail and systems modeling add value?

## Scope

- Genome/epigenome, signaling, metabolism, proteostasis, cell fate, matrix, tissues, microbiome, development and systems regulation.
- Cross-cutting molecular and cellular failure modes.
- Genomic, omic, spatial, tissue, cellular, model and perturbation measurement.
- Molecular, cellular, gene, regenerative and prevention interventions.

## Reading conventions

- **Established:** replicated mechanisms, mature measurement standards, or interventions supported by convergent evidence.
- **Context-limited:** useful under defined populations, settings, reference methods, or intended uses; transportability must be demonstrated.
- **Frontier:** biologically plausible or technically promising but not yet validated for broad clinical claims.
- **Measurement claim:** always interpreted as an inference from a signal chain rather than a direct view of the latent biological state.

# Contents

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|                                                                                                                                                                   |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Document status and intended use</b>                                                                                                                           | <b>i</b>  |
| <b>How to use this volume</b>                                                                                                                                     | <b>ii</b> |
| <b>1 Executive synthesis</b>                                                                                                                                      | <b>1</b>  |
| 1.1 What a complete profile requires . . . . .                                                                                                                    | 1         |
| <b>2 Historical development and changing models</b>                                                                                                               | <b>3</b>  |
| 2.0.1 Living processes were explained through vital, elemental and humoral systems. . . . .                                                                       | 3         |
| 2.0.2 Cells, tissues, heredity and pathology established structural biological units. . . . .                                                                     | 4         |
| 2.0.3 Biochemistry, physiology, microbiology, genetics and molecular biology revealed pathways and information flow. . . . .                                      | 4         |
| 2.0.4 Cell signaling, immunology, developmental biology, randomized evidence and biotechnology linked mechanisms to interventions. . . . .                        | 4         |
| 2.0.5 Genomics, single-cell/spatial omics, CRISPR, organoids and computational systems biology expanded resolution. . . . .                                       | 4         |
| 2.0.6 Current biology integrates molecules, cells, mechanics, ecology, development, behavior and environment with explicit measurement and causal models. . . . . | 4         |
| <b>3 Normal foundations: structure, physiology, and regulation</b>                                                                                                | <b>5</b>  |
| 3.0.1 Genome and inherited variation . . . . .                                                                                                                    | 5         |
| 3.0.2 Epigenome and gene regulation . . . . .                                                                                                                     | 5         |
| 3.0.3 Cell membranes, receptors and signaling . . . . .                                                                                                           | 5         |
| 3.0.4 Energy metabolism and mitochondria . . . . .                                                                                                                | 6         |
| 3.0.5 Protein synthesis, folding and turnover . . . . .                                                                                                           | 6         |
| 3.0.6 Cell cycle, senescence and death . . . . .                                                                                                                  | 6         |
| 3.0.7 Extracellular matrix and mechanobiology . . . . .                                                                                                           | 7         |
| 3.0.8 Cell-cell communication and tissue organization . . . . .                                                                                                   | 7         |
| 3.0.9 Stem cells, differentiation and regeneration . . . . .                                                                                                      | 7         |
| 3.0.10 Microbiome and host ecology . . . . .                                                                                                                      | 8         |
| 3.0.11 Developmental programs and plasticity . . . . .                                                                                                            | 8         |
| 3.0.12 Systems regulation and emergent phenotype . . . . .                                                                                                        | 9         |
| 3.1 Useful quantitative relationships . . . . .                                                                                                                   | 9         |
| <b>4 Determinants, heterogeneity, and causal architecture</b>                                                                                                     | <b>10</b> |
| 4.0.1 Genetic architecture and family history . . . . .                                                                                                           | 12        |
| 4.0.2 Development and life course . . . . .                                                                                                                       | 12        |
| 4.0.3 Nutrition and metabolism . . . . .                                                                                                                          | 12        |

|          |                                                                            |           |
|----------|----------------------------------------------------------------------------|-----------|
| 4.0.4    | Physical activity, conditioning, and sedentary behavior . . . . .          | 13        |
| 4.0.5    | Sleep, circadian timing, and recovery . . . . .                            | 13        |
| 4.0.6    | Stress, emotion, and behavior . . . . .                                    | 13        |
| 4.0.7    | Social, economic, cultural, and care conditions . . . . .                  | 14        |
| 4.0.8    | Environmental and occupational exposures . . . . .                         | 14        |
| 4.0.9    | Health care, medication, and iatrogenic effects . . . . .                  | 14        |
| <b>5</b> | <b>Major disorders, failure modes, and clinically important phenotypes</b> | <b>15</b> |
| 5.0.1    | Pathogenic inherited variation . . . . .                                   | 15        |
| 5.0.2    | Somatic mutation and clonal evolution . . . . .                            | 15        |
| 5.0.3    | Epigenetic dysregulation . . . . .                                         | 16        |
| 5.0.4    | Protein misfolding and aggregation . . . . .                               | 16        |
| 5.0.5    | Mitochondrial and bioenergetic failure . . . . .                           | 17        |
| 5.0.6    | Oxidative and reductive stress . . . . .                                   | 17        |
| 5.0.7    | Cellular senescence and impaired regeneration . . . . .                    | 18        |
| 5.0.8    | Dysregulated cell death and inflammation . . . . .                         | 18        |
| 5.0.9    | Fibrotic remodeling . . . . .                                              | 19        |
| 5.0.10   | Barrier and transport failure . . . . .                                    | 19        |
| 5.0.11   | Endocrine/receptor resistance and signal mismatch . . . . .                | 20        |
| 5.0.12   | Immune dysregulation and loss of tolerance . . . . .                       | 20        |
| 5.0.13   | Developmental programming and constrained reserve . . . . .                | 20        |
| 5.0.14   | Dysbiosis and ecological disruption . . . . .                              | 21        |
| <b>6</b> | <b>Assessment, clinical reasoning, and longitudinal profiling</b>          | <b>22</b> |
| 6.0.1    | 1. Question and urgency . . . . .                                          | 23        |
| 6.0.2    | 2. History and lived phenotype . . . . .                                   | 23        |
| 6.0.3    | 3. Examination and basic measurements . . . . .                            | 23        |
| 6.0.4    | 4. Targeted laboratory, imaging, and physiology . . . . .                  | 24        |
| 6.0.5    | 5. Function and challenge testing . . . . .                                | 24        |
| 6.0.6    | 6. Longitudinal integration . . . . .                                      | 24        |
| <b>7</b> | <b>Measurement and device landscape</b>                                    | <b>25</b> |
| 7.0.1    | Karyotyping and cytogenetics . . . . .                                     | 27        |
| 7.0.2    | DNA sequencing and genotyping . . . . .                                    | 27        |
| 7.0.3    | Transcriptomics and epigenomics . . . . .                                  | 27        |
| 7.0.4    | Proteomics and functional protein assays . . . . .                         | 28        |
| 7.0.5    | Metabolomics and flux analysis . . . . .                                   | 28        |
| 7.0.6    | Single-cell and spatial omics . . . . .                                    | 29        |
| 7.0.7    | Histology and molecular pathology . . . . .                                | 29        |
| 7.0.8    | Cellular functional assays and electrophysiology . . . . .                 | 29        |

|           |                                                                       |           |
|-----------|-----------------------------------------------------------------------|-----------|
| 7.0.9     | Organoids and induced pluripotent-cell models . . . . .               | 30        |
| 7.0.10    | Animal and comparative models . . . . .                               | 30        |
| 7.0.11    | Molecular imaging and tracers . . . . .                               | 30        |
| 7.0.12    | Perturbation and challenge experiments . . . . .                      | 31        |
| 7.0.13    | Multi-omics integration and systems models . . . . .                  | 31        |
| <b>8</b>  | <b>Engineering architectures, signal quality, and validation</b>      | <b>32</b> |
| 8.0.1     | Intended use and measurand . . . . .                                  | 33        |
| 8.0.2     | Sensor and transduction . . . . .                                     | 33        |
| 8.0.3     | Timing and synchronization . . . . .                                  | 33        |
| 8.0.4     | Analog, acquisition, and digital integrity . . . . .                  | 34        |
| 8.0.5     | Calibration and traceability . . . . .                                | 34        |
| 8.0.6     | Artifact and quality channels . . . . .                               | 34        |
| 8.0.7     | Algorithms and uncertainty . . . . .                                  | 35        |
| 8.0.8     | Safety, usability, cybersecurity, and privacy . . . . .               | 35        |
| <b>9</b>  | <b>Prevention, treatment, rehabilitation, and system intervention</b> | <b>36</b> |
| 9.0.1     | Exposure and developmental prevention . . . . .                       | 37        |
| 9.0.2     | Nutrient and cofactor replacement . . . . .                           | 38        |
| 9.0.3     | Small-molecule pathway modulation . . . . .                           | 38        |
| 9.0.4     | Biologic and antibody therapy . . . . .                               | 38        |
| 9.0.5     | Hormone and physiologic replacement . . . . .                         | 38        |
| 9.0.6     | Cellular therapy and transplantation . . . . .                        | 39        |
| 9.0.7     | Gene addition, editing and RNA therapy . . . . .                      | 39        |
| 9.0.8     | Proteostasis and degradation therapy . . . . .                        | 39        |
| 9.0.9     | Regenerative and tissue-engineering strategies . . . . .              | 39        |
| 9.0.10    | Microbiome-directed intervention . . . . .                            | 40        |
| 9.0.11    | Senescence/aging-targeted intervention . . . . .                      | 40        |
| 9.0.12    | Mechanism-informed combination therapy . . . . .                      | 40        |
| <b>10</b> | <b>What is known, not known, and frequently overstated</b>            | <b>41</b> |
| 10.0.1    | Can biological age be measured meaningfully? . . . . .                | 42        |
| 10.0.2    | Can multi-omics define actionable subtypes? . . . . .                 | 42        |
| 10.0.3    | How much risk is genetic versus environmental? . . . . .              | 43        |
| 10.0.4    | Can one composite score summarize health? . . . . .                   | 43        |
| 10.0.5    | When is earlier detection beneficial? . . . . .                       | 43        |
| 10.0.6    | Do digital biomarkers generalize? . . . . .                           | 44        |
| 10.0.7    | What counts as meaningful change? . . . . .                           | 44        |
| <b>11</b> | <b>Research and development roadmap</b>                               | <b>45</b> |
| 11.0.1    | 0. Cross-scale perturbation atlas . . . . .                           | 46        |

|                                                                |           |
|----------------------------------------------------------------|-----------|
| 11.0.2 0. Translation failure registry . . . . .               | 46        |
| 11.0.3 1. Reference map . . . . .                              | 46        |
| 11.0.4 2. Causal determinants . . . . .                        | 47        |
| 11.0.5 3. Measurement validity . . . . .                       | 47        |
| 11.0.6 4. Clinical utility . . . . .                           | 47        |
| 11.0.7 5. Longitudinal multimorbidity . . . . .                | 48        |
| 11.0.8 6. Implementation and governance . . . . .              | 48        |
| <b>12 Cross-volume integration</b>                             | <b>49</b> |
| <b>A Minimum data schema for a longitudinal domain profile</b> | <b>50</b> |
| <b>B Glossary</b>                                              | <b>51</b> |
| <b>Selected references and current standards</b>               | <b>52</b> |

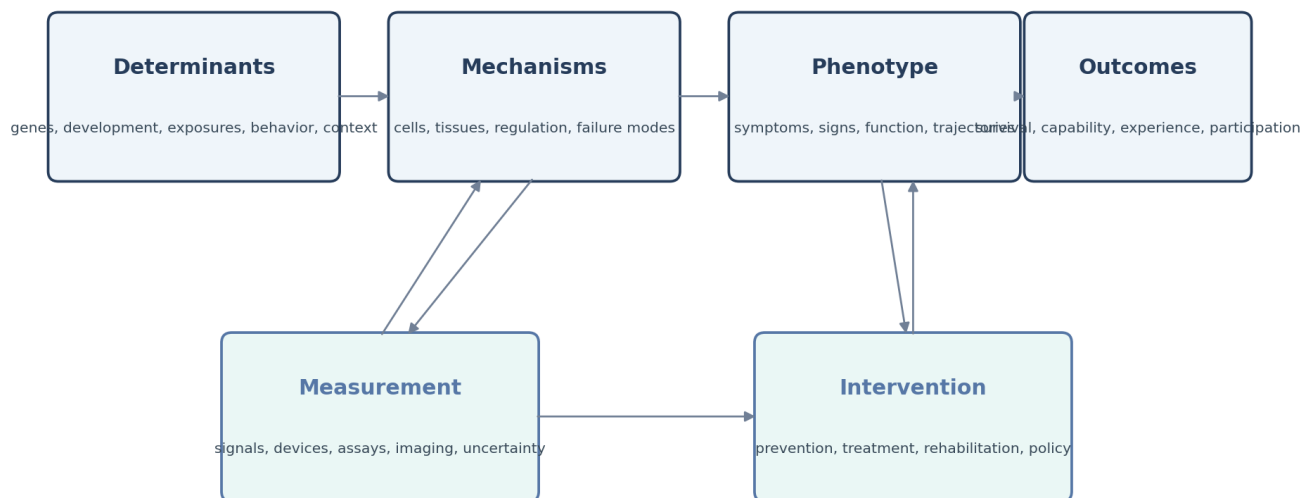
# Executive synthesis

Human biology is a nested adaptive system: sequence and regulation shape cells; cells form tissues; tissues exchange matter, force and signals; organs and behavior operate within social and physical environments. No molecular layer is inherently more “real” than function, and multi-omics does not create causality by volume. Useful translation requires an explicit chain from mechanism to measurable state to decision and patient-important outcome.

## Central organizing principle

Provide the shared mechanistic substrate beneath organ systems while distinguishing causal biology from high-dimensional correlation and molecular marketing.

## Biological Foundations and Regulation



A health profile is a causal, temporal, functional, and measurement model - not a single score.

**Figure 1.1:** The causal and measurement architecture used throughout this volume.

## 1.1 What a complete profile requires

- A multiscale account from molecules and cells through tissues, organs, whole-person function, and context.
- A longitudinal account of baseline, reserve under challenge, variability, compensation, injury, recovery, and irreversible change.
- Separation of causal determinants from correlates, biomarkers, manifestations, and consequences.

- Explicit representation of measurement error, missingness, artifact, reference methods, population drift, and clinical actionability.
- Integration with other atlas domains rather than treating one specialty as a closed system.

**Safety boundary**

Genomic and omic results can expose familial risk, identity and incidental findings. Experimental gene, cell and regenerative interventions can produce irreversible and delayed harm and require validated manufacturing, consent and long-term oversight.

# Historical development and changing models

The history of a domain is not decorative. It explains why present-day categories, instruments, thresholds, specialties, and blind spots look the way they do. Technical advances often changed the object of medicine itself: what could be seen, counted, localized, or treated became the preferred unit of explanation.

| Period                                       | Development                                                                                                                                         | What it made possible                                        | Residual limitation                                                                                |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Early holistic and moral models              | Living processes were explained through vital, elemental and humoral systems.                                                                       | Recognized integration and change.                           | Mechanisms were not experimentally separable.                                                      |
| Anatomical and pathological organization     | Cells, tissues, heredity and pathology established structural biological units.                                                                     | Made disease localization and experimental biology possible. | Static structure underrepresented dynamics and environment.                                        |
| Statistical and laboratory medicine          | Biochemistry, physiology, microbiology, genetics and molecular biology revealed pathways and information flow.                                      | Enabled targeted measurement and therapy.                    | Linear pathway diagrams obscured networks and heterogeneity.                                       |
| Biopsychosocial and evidence-based expansion | Cell signaling, immunology, developmental biology, randomized evidence and biotechnology linked mechanisms to interventions.                        | Produced high-impact drugs and diagnostics.                  | Mechanistic plausibility was sometimes mistaken for clinical efficacy.                             |
| Systems, data and precision era              | Genomics, single-cell/spatial omics, CRISPR, organoids and computational systems biology expanded resolution.                                       | Allows multiscale maps and perturbation.                     | Batch effects, causal ambiguity and translation gaps scale with data volume.                       |
| Current integrated model                     | Current biology integrates molecules, cells, mechanics, ecology, development, behavior and environment with explicit measurement and causal models. | Provides a shared substrate for the atlas.                   | Institutional and commercial incentives still favor isolated biomarkers and deterministic stories. |

## 2.0.1 Living processes were explained through vital, elemental and humoral systems.

**Scientific consequence.** Recognized integration and change.

**What remained unresolved.** Mechanisms were not experimentally separable.

*Evidence anchors:* <sup>[3,4]</sup>

## 2.0.2 Cells, tissues, heredity and pathology established structural biological units.

**Scientific consequence.** Made disease localization and experimental biology possible.

**What remained unresolved.** Static structure underrepresented dynamics and environment.

*Evidence anchors:* <sup>[3,4]</sup>

## 2.0.3 Biochemistry, physiology, microbiology, genetics and molecular biology revealed pathways and information flow.

**Scientific consequence.** Enabled targeted measurement and therapy.

**What remained unresolved.** Linear pathway diagrams obscured networks and heterogeneity.

*Evidence anchors:* <sup>[3,4]</sup>

## 2.0.4 Cell signaling, immunology, developmental biology, randomized evidence and biotechnology linked mechanisms to interventions.

**Scientific consequence.** Produced high-impact drugs and diagnostics.

**What remained unresolved.** Mechanistic plausibility was sometimes mistaken for clinical efficacy.

*Evidence anchors:* <sup>[3,4]</sup>

## 2.0.5 Genomics, single-cell/spatial omics, CRISPR, organoids and computational systems biology expanded resolution.

**Scientific consequence.** Allows multiscale maps and perturbation.

**What remained unresolved.** Batch effects, causal ambiguity and translation gaps scale with data volume.

*Evidence anchors:* <sup>[3,4]</sup>

## 2.0.6 Current biology integrates molecules, cells, mechanics, ecology, development, behavior and environment with explicit measurement and causal models.

**Scientific consequence.** Provides a shared substrate for the atlas.

**What remained unresolved.** Institutional and commercial incentives still favor isolated biomarkers and deterministic stories.

*Evidence anchors:* <sup>[1,11]</sup>

# Normal foundations: structure, physiology, and regulation

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Normal physiology is represented here as regulated ranges and adaptive control, not as one fixed number. Many variables are intentionally dynamic, tissue-specific, state-dependent, and coupled to other systems. Disease can arise from primary damage, maladaptive compensation, loss of reserve, mistimed regulation, or mismatch between organism and environment.

## 3.0.1 Genome and inherited variation

**Core principle.** DNA sequence and chromosome structure provide heritable instructions and variation affecting development and susceptibility.

**Control architecture.** Replication/repair, gene regulation, meiosis, recombination and population history.

**Failure modes.** Pathogenic variants, aneuploidy, repeat expansion, mosaicism or altered susceptibility.

**Observable consequences.** Karyotype, sequencing, copy-number and family/phenotype evidence.

**Cross-system coupling.** Development, reproduction, cancer and all organ systems.

*Evidence anchors:* [3,4,8]

## 3.0.2 Epigenome and gene regulation

**Core principle.** Chromatin state, DNA modification, transcription factors and noncoding regulation control context-specific gene expression.

**Control architecture.** Development, cell identity, hormones, environment, aging and feedback.

**Failure modes.** Misexpression, imprinting disorders, unstable cell identity or altered stress response.

**Observable consequences.** Epigenomic/transcriptomic assays with cell-type and temporal context.

**Cross-system coupling.** Development, cancer, environment and aging.

*Evidence anchors:* [3,7,9]

## 3.0.3 Cell membranes, receptors and signaling

**Core principle.** Cells sense ligands, voltage, force and nutrients through receptors and intracellular networks.

**Control architecture.** Binding, second messengers, phosphorylation, trafficking, feedback and degradation.

**Failure modes.** Loss/gain of signaling, desensitization, channelopathy, endocrine resistance or oncogenic activation.

**Observable consequences.** Functional assays, electrophysiology, biomarkers and molecular tests.

**Cross-system coupling.** Endocrine, nervous, immune, cardiovascular and pharmacology.

*Evidence anchors:* [3,5,16]

### 3.0.4 Energy metabolism and mitochondria

**Core principle.** Cells convert nutrients and oxygen to ATP and biosynthetic intermediates while managing redox state.

**Control architecture.** Glycolysis, TCA/oxidative phosphorylation, substrate switching, organelle quality control and signaling.

**Failure modes.** Energy failure, oxidative injury, metabolic inflexibility, mitochondrial disease or apoptosis.

**Observable consequences.** Metabolites, flux, oxygen consumption, imaging and genetics.

**Cross-system coupling.** Muscle, nervous, endocrine, aging and critical illness.

*Evidence anchors:* [3,5,9]

### 3.0.5 Protein synthesis, folding and turnover

**Core principle.** Proteins are translated, modified, trafficked, assembled and removed through proteostasis systems.

**Control architecture.** Ribosome/ER/Golgi, chaperones, ubiquitin-proteasome, autophagy and stress responses.

**Failure modes.** Misfolding/aggregation, enzyme/receptor deficiency, ER stress or toxic gain of function.

**Observable consequences.** Proteomics, activity assays, pathology, imaging and genetic evidence.

**Cross-system coupling.** Neurologic, liver, muscle, immune and aging.

*Evidence anchors:* [3,4,9]

### 3.0.6 Cell cycle, senescence and death

**Core principle.** Proliferation, quiescence, senescence and regulated death balance growth, repair and tumor prevention.

**Control architecture.** Checkpoints, telomeres, DNA damage, apoptosis, necroptosis/pyroptosis and immune clearance.

**Failure modes.** Cancer, tissue depletion, chronic senescence/inflammation or excessive cell death.

**Observable consequences.** Histology, proliferation/death markers, clonality and functional outcomes.

**Cross-system coupling.** Cancer, aging, immune and repair systems.

*Evidence anchors:* [4,9,10]

### 3.0.7 Extracellular matrix and mechanobiology

**Core principle.** Cells and matrix create tissues whose force, geometry and stiffness feed back on cell behavior.

**Control architecture.** Integrins, cytoskeleton, mechanical stress, matrix deposition/remodeling and fluid forces.

**Failure modes.** Fibrosis, fragility, altered development, tumor invasion or organ stiffness.

**Observable consequences.** Biomechanics, imaging, histology and matrix biomarkers.

**Cross-system coupling.** Musculoskeletal, cardiovascular, lung, liver, skin and cancer.

*Evidence anchors:* [4,11]

### 3.0.8 Cell-cell communication and tissue organization

**Core principle.** Direct contact, paracrine/endocrine signals, extracellular vesicles and neural/vascular networks coordinate cells in space and time.

**Control architecture.** Developmental patterning, polarity, junctions, morphogens, perfusion and immune surveillance.

**Failure modes.** Barrier failure, disordered architecture, dysplasia, inflammation or developmental anomaly.

**Observable consequences.** Histology, spatial omics, imaging and functional assays.

**Cross-system coupling.** All organs, development and cancer.

*Evidence anchors:* [3,4,6]

### 3.0.9 Stem cells, differentiation and regeneration

**Core principle.** Stem/progenitor cells self-renew and generate specialized lineages with tissue-specific regenerative capacity.

**Control architecture.** Niche, transcriptional networks, injury signals, age and immune/metabolic context.

**Failure modes.** Developmental failure, marrow/tissue failure, maladaptive repair or tumor stemness.

**Observable consequences.** Lineage tracing in models, single-cell/spatial profiles and clinical regeneration outcomes.

**Cross-system coupling.** Hematologic, skin, GI, reproductive, cancer and aging.

*Evidence anchors:* [3,6,9]

### 3.0.10 Microbiome and host ecology

**Core principle.** Microbial communities interact with barriers, immunity, metabolism and exposures without being simply good or bad.

**Control architecture.** Colonization resistance, metabolites, immune education, diet/drugs and ecological disturbance.

**Failure modes.** Infection, dysbiosis association, altered metabolism or immune signaling.

**Observable consequences.** Culture, sequencing/metagenomics/metabolomics and causal intervention response.

**Cross-system coupling.** GI, skin, oral, reproductive, immune and One Health.

*Evidence anchors:* [7,14,15]

### 3.0.11 Developmental programs and plasticity

**Core principle.** Organism and organs form through timed gene-environment interactions, with sensitive periods and adaptive plasticity.

**Control architecture.** Cell fate, morphogenesis, maternal/placental signals, nutrition, hormones and experience.

**Failure modes.** Congenital anomaly, altered reserve, developmental delay or later disease susceptibility.

**Observable consequences.** Prenatal history, growth, milestones, imaging/genetics and longitudinal function.

**Cross-system coupling.** Maternal, life course, nervous and all organs.

*Evidence anchors:* [3,12,13]

### 3.0.12 Systems regulation and emergent phenotype

**Core principle.** Phenotype emerges from networks with feedback, redundancy, thresholds and multi-scale interactions.

**Control architecture.** Control loops, network topology, adaptation, stochasticity and environment.

**Failure modes.** Nonlinear transitions, compensation, oscillation, fragility and heterogeneous disease.

**Observable consequences.** Multimodal longitudinal data, perturbation, mechanistic models and functional outcome.

**Cross-system coupling.** Homeostasis, measurement and precision medicine.

*Evidence anchors:* [8,11,24]

## 3.1 Useful quantitative relationships

These equations are compact models. Their variables require operational definitions, units, boundary conditions, and a statement of when the approximation fails.

| Relationship               | Expression                                             | Interpretation                                                    | Limits                                                                                            |
|----------------------------|--------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Michaelis-Menten kinetics  | $v = \frac{V_{max}[S]}{K_m + [S]}$                     | Idealized saturable enzyme or transport behavior.                 | Assumes simplified steady state and may fail in networks, compartments or allostery.              |
| Hill relationship          | $E = \frac{E_{max}C^n}{EC_{50}^n + C^n}$               | Represents cooperative or steep concentration-response behavior.  | Phenomenologic; n need not equal binding stoichiometry and exposure at the target may be unknown. |
| Hardy-Weinberg expectation | $p^2 + 2pq + q^2 = 1$                                  | Baseline genotype frequencies under ideal population assumptions. | Selection, structure, assortative mating, drift and genotyping error violate assumptions.         |
| Nernst potential           | $E = \frac{RT}{zF} \ln \frac{[ion]_{out}}{[ion]_{in}}$ | Equilibrium potential for one ion across a membrane.              | Real membranes have multiple permeabilities, pumps and non-equilibrium dynamics.                  |

# Determinants, heterogeneity, and causal architecture

No universal percentage can divide health into genetics and environment. Contributions vary by outcome, age, population, exposure distribution, measurement scale, and causal estimand. Genetic liability can alter sensitivity to an exposure; exposures can alter gene regulation; both operate through development, behavior, social conditions, and access to prevention or care.

| Determinant                             | Main pathways                                                                                                                                                                                                                                              | Leverage                                                                                                                                               | Measurement                                                                                                                                                                          | Interpretive caution                                                                                                                                                    |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Genetic architecture and family history | Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within biological foundations and regulation. | Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.                   | Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question. | Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses. |
| Development and life course             | Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.                                                                            | Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories. | Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.                                        | Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.                                        |
| Nutrition and metabolism                | Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.                                                                | Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.        | Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.                                                   | Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.                                                         |

Continued on next page

| <b>Determinant</b>                                      | <b>Main pathways</b>                                                                                                                                                            | <b>Leverage</b>                                                                                                                                                  | <b>Measurement</b>                                                                                                                                           | <b>Interpretive caution</b>                                                                                                                   |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Physical activity, conditioning, and sedentary behavior | Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.                                        | Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.                                     | Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.                                            | Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.                     |
| Sleep, circadian timing, and recovery                   | Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.               | Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.                 | History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.                                                        | Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.                |
| Stress, emotion, and behavior                           | Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology. | Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors. | Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score. | Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.                                  |
| Social, economic, cultural, and care conditions         | Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.                | Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.                              | Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.                                 | Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.                       |
| Environmental and occupational exposures                | Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.          | Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.                                                | Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.                              | Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably. |

Continued on next page

| Determinant                                     | Main pathways                                                                                                                                                     | Leverage                                                                                                                         | Measurement                                                                                                                 | Interpretive caution                                                                                                       |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Health care, medication, and iatrogenic effects | Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm. | Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up. | Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals. | Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease. |

### 4.0.1 Genetic architecture and family history

**Mechanistic pathways.** Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within biological foundations and regulation.

**Potentially modifiable components.** Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.

**How to measure it.** Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.

**Causal and equity cautions.** Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.

*Evidence anchors:* [3,8]

### 4.0.2 Development and life course

**Mechanistic pathways.** Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.

**Potentially modifiable components.** Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.

**How to measure it.** Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.

**Causal and equity cautions.** Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.

*Evidence anchors:* [12,13]

### 4.0.3 Nutrition and metabolism

**Mechanistic pathways.** Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.

**Potentially modifiable components.** Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.

**How to measure it.** Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.

**Causal and equity cautions.** Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

#### 4.0.4 Physical activity, conditioning, and sedentary behavior

**Mechanistic pathways.** Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.

**Potentially modifiable components.** Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.

**How to measure it.** Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.

**Causal and equity cautions.** Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

#### 4.0.5 Sleep, circadian timing, and recovery

**Mechanistic pathways.** Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.

**Potentially modifiable components.** Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.

**How to measure it.** History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.

**Causal and equity cautions.** Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.

#### 4.0.6 Stress, emotion, and behavior

**Mechanistic pathways.** Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.

**Potentially modifiable components.** Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.

**How to measure it.** Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.

**Causal and equity cautions.** Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.

#### 4.0.7 Social, economic, cultural, and care conditions

**Mechanistic pathways.** Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.

**Potentially modifiable components.** Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.

**How to measure it.** Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.

**Causal and equity cautions.** Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.

#### 4.0.8 Environmental and occupational exposures

**Mechanistic pathways.** Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.

**Potentially modifiable components.** Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.

**How to measure it.** Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.

**Causal and equity cautions.** Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

*Evidence anchors:* <sup>[21]</sup>

#### 4.0.9 Health care, medication, and iatrogenic effects

**Mechanistic pathways.** Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.

**Potentially modifiable components.** Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.

**How to measure it.** Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.

**Causal and equity cautions.** Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

# Major disorders, failure modes, and clinically important phenotypes

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The profiles below are not a replacement for a diagnostic manual. They are organized to connect mechanism, phenotype, measurement, intervention, and uncertainty. The primary unit is the biological process, molecular/cellular failure mode, or multiscale mechanism, but cross-system processes and functional effects are retained.

## 5.0.1 Pathogenic inherited variation

**Definition and phenotype.** A variant or chromosome change with sufficient evidence of causal contribution to a phenotype.

**Pathophysiology.** Loss/gain of function, dosage, dominant-negative, toxic RNA/protein or altered regulation can disrupt specific tissues.

**Determinants and modifiers.** Inheritance, de novo event, mosaicism, modifier genes and environment.

**Assessment.** Phenotype-first genetics, family segregation, population frequency, functional and curated evidence.

**Intervention logic.** Prevention/surveillance, replacement/targeted therapy, reproductive counseling or gene/cell approaches where valid.

**Known limits and open questions.** Penetrance and expressivity vary; a variant classification is not a complete prognosis.

**Suite links.** 07 life course; 15 genomic ethics; organ volumes.

*Evidence anchors:* <sup>[3,8,30]</sup>

## 5.0.2 Somatic mutation and clonal evolution

**Definition and phenotype.** Acquired variants create cell clones that may remain benign, alter function or progress to cancer.

**Pathophysiology.** Mutation and selection under tissue, immune, age and treatment pressures reshape populations.

**Determinants and modifiers.** Age, replication, inflammation, radiation/chemicals, infection and therapy.

**Assessment.** Tissue/blood sequencing, clonality, histology and longitudinal burden.

**Intervention logic.** Exposure reduction where known, surveillance or clone-/cancer-directed treatment by risk.

**Known limits and open questions.** Low-level clones are common and action thresholds are often uncertain.

**Suite links.** 04 cancer; 03G hematology; 07 aging.

*Evidence anchors:* [9,10,20]

### 5.0.3 Epigenetic dysregulation

**Definition and phenotype.** Abnormal chromatin or imprinting state changes gene expression without changing primary sequence.

**Pathophysiology.** Developmental error, mutation of chromatin machinery, environment or aging alters regulatory programs.

**Determinants and modifiers.** Inherited machinery variants, tumors, prenatal exposures and stochastic drift.

**Assessment.** Cell-type-specific methylation/chromatin/transcript tests plus phenotype.

**Intervention logic.** Cause-specific care; epigenetic drugs in selected malignancy; exposure prevention where causal.

**Known limits and open questions.** Many epigenetic marks are consequences or correlates rather than modifiable causes.

**Suite links.** 09 exposome; 04 cancer; 07 development.

*Evidence anchors:* [7,9,23]

### 5.0.4 Protein misfolding and aggregation

**Definition and phenotype.** Proteins adopt abnormal conformations or accumulate, impairing cellular function or spreading pathology.

**Pathophysiology.** Mutation, aging, oxidative/ER stress and impaired degradation exceed proteostasis.

**Determinants and modifiers.** Genetics, age, environment, metabolic stress and tissue-specific vulnerability.

**Assessment.** Pathology, fluid/imaging biomarkers, genetics and functional phenotype.

**Intervention logic.** Reduce production/aggregation, enhance clearance or replace function in disease-specific ways.

**Known limits and open questions.** Biomarker positivity may precede symptoms for years and treatment benefit can be limited.

**Suite links.** 03C nervous; 04 degeneration; 14 biologics.

*Evidence anchors:* [3,9]

### 5.0.5 Mitochondrial and bioenergetic failure

**Definition and phenotype.** Insufficient or dysregulated energy production and organelle signaling across inherited or acquired states.

**Pathophysiology.** mtDNA/nuclear defects, toxin/drug, hypoxia, aging or secondary disease impair ATP/redox/calcium control.

**Determinants and modifiers.** Genetics, infection, medications, nutrition and organ stress.

**Assessment.** Lactate/metabolites, respiratory-chain assays, genetics, biopsy/imaging and exercise function.

**Intervention logic.** Treat specific deficiencies/toxins, supportive organ care and selected targeted/gene approaches.

**Known limits and open questions.** Nonspecific “mitochondrial dysfunction” is often overclaimed from weak assays.

**Suite links.** 03O exercise; 03C/I; 14 therapeutics.

*Evidence anchors:* [3,8,25]

### 5.0.6 Oxidative and reductive stress

**Definition and phenotype.** Reactive species or altered redox balance disrupts lipids, proteins, DNA and signaling.

**Pathophysiology.** Production from metabolism/inflammation/toxins exceeds compartment-specific defense and repair.

**Determinants and modifiers.** Ischemia, infection, pollution, smoking, metabolic disease and radiation.

**Assessment.** Damage/adduct biomarkers, redox assays and tissue/organ outcomes.

**Intervention logic.** Remove causal exposure and treat disease; broad antioxidant supplementation rarely substitutes for mechanism.

**Known limits and open questions.** Markers are unstable/nonspecific and antioxidants can disrupt signaling or harm.

**Suite links.** 09 environment; 04 ischemia/inflammation.

*Evidence anchors:* [4,21,23]

### 5.0.7 Cellular senescence and impaired regeneration

**Definition and phenotype.** Persistent cell-cycle arrest with altered secretory/signaling state can limit repair and promote inflammation.

**Pathophysiology.** Telomere/DNA damage, oncogenic and metabolic stress induce heterogeneous senescent states.

**Determinants and modifiers.** Age, radiation, chemotherapy, chronic disease and tissue injury.

**Assessment.** Research panels of markers plus tissue/function; no universal clinical senescence assay.

**Intervention logic.** Risk-factor reduction and disease care; senolytic/senomorphing strategies remain mostly experimental.

**Known limits and open questions.** Marker specificity, beneficial transient senescence and long-term safety are unresolved.

**Suite links.** 07 aging; 14 frontier therapy.

*Evidence anchors:* [9,25]

### 5.0.8 Dysregulated cell death and inflammation

**Definition and phenotype.** Excess apoptosis/necrosis/pyroptosis or failed clearance causes tissue loss and inflammatory damage.

**Pathophysiology.** Energy failure, toxins, immune pathways and damage sensing activate distinct death programs.

**Determinants and modifiers.** Ischemia, infection, autoimmunity, neurodegeneration and therapy.

**Assessment.** Tissue pathology, circulating injury markers, imaging and organ function.

**Intervention logic.** Remove cause and modulate pathway only where disease-specific evidence exists.

**Known limits and open questions.** Death markers overlap and blocking one pathway may redirect another.

**Suite links.** 04 injury/sepsis; organ volumes.

*Evidence anchors:* <sup>[4]</sup>

### 5.0.9 Fibrotic remodeling

**Definition and phenotype.** Persistent repair programs deposit and crosslink extracellular matrix, distorting organ architecture.

**Pathophysiology.** Repeated injury, immune signals, fibroblast activation and mechanical feedback sustain scarring.

**Determinants and modifiers.** Chronic infection/inflammation, toxins, metabolic disease, radiation and genetics.

**Assessment.** Imaging/stiffness, organ function, biomarkers and tissue pathology.

**Intervention logic.** Remove injury, disease-specific antifibrotic/immune therapy and manage organ failure.

**Known limits and open questions.** Biomarkers and stiffness are not equivalent to reversible active fibrosis.

**Suite links.** 04 fibrosis; lung/liver/kidney/heart/skin.

*Evidence anchors:* <sup>[4,11]</sup>

### 5.0.10 Barrier and transport failure

**Definition and phenotype.** Epithelial/endothelial barriers or membrane transporters fail to maintain selective exchange.

**Pathophysiology.** Junction, channel/transporter, inflammation, ischemia or toxins alter permeability and gradients.

**Determinants and modifiers.** Genetics, infection, drugs, immune and environmental injury.

**Assessment.** Flux, permeability, electrophysiology, imaging, tissue and systemic consequences.

**Intervention logic.** Restore cause/transport and protect fluid/nutrition/infection state.

**Known limits and open questions.** In vitro barrier measures may not translate to in vivo organ function.

**Suite links.** 03J skin; 03E GI; 03F kidney; 03B lung.

*Evidence anchors:* <sup>[3,4]</sup>

### 5.0.11 Endocrine/receptor resistance and signal mismatch

**Definition and phenotype.** Signal production, delivery, receptor response or timing fails despite measurable hormone/ligand levels.

**Pathophysiology.** Receptor/postreceptor defects, binding/clearance, feedback adaptation or mistimed rhythms.

**Determinants and modifiers.** Genetics, obesity/metabolism, drugs, autoimmunity and tumors.

**Assessment.** Dynamic testing, free/total hormone context, receptor/genetic studies selectively and target-organ function.

**Intervention logic.** Replace or suppress signal, improve sensitivity or target downstream pathway by disease.

**Known limits and open questions.** A single concentration cannot establish tissue action.

**Suite links.** 03D endocrine; 03O circadian; 14 pharmacology.

*Evidence anchors:* <sup>[16,17]</sup>

### 5.0.12 Immune dysregulation and loss of tolerance

**Definition and phenotype.** Protective immunity becomes deficient, persistent, misdirected or unable to resolve.

**Pathophysiology.** Innate/adaptive signaling, barrier, antigen, tolerance and metabolic context interact.

**Determinants and modifiers.** Genetics, infection, microbiome, environment, aging and therapy.

**Assessment.** Cellular/functional/antibody/tissue tests with clinical phenotype.

**Intervention logic.** Vaccination/replacement, source control or targeted immunomodulation.

**Known limits and open questions.** Cytokines and autoantibodies are often nonspecific.

**Suite links.** 03H immune; 08 infection; 04 inflammation.

*Evidence anchors:* <sup>[18,19]</sup>

### 5.0.13 Developmental programming and constrained reserve

**Definition and phenotype.** Early conditions alter organ structure, regulation or capacity with later consequences.

**Pathophysiology.** Sensitive-period nutrition, stress, toxins, placenta, infection and experience influence development and epigenetic regulation.

**Determinants and modifiers.** Maternal/fetal health, poverty, environment, genetics and care.

**Assessment.** Life-course history, growth/development, organ reserve and longitudinal cohorts.

**Intervention logic.** Preconception/early prevention, supportive development and later risk reduction.

**Known limits and open questions.** Retrospective attribution is vulnerable to confounding and determinism.

**Suite links.** 03N maternal; 07 life course; 09 environment.

*Evidence anchors:* [12,13,22]

#### 5.0.14 Dysbiosis and ecological disruption

**Definition and phenotype.** A microbial community state associated with impaired host function or disease under a defined site/context.

**Pathophysiology.** Antibiotics, diet, infection, immune/barrier state and environment alter ecological networks and metabolites.

**Determinants and modifiers.** Drugs, diet, geography, age, disease and sampling.

**Assessment.** Site-specific sequencing/culture/metabolites plus host phenotype and intervention response.

**Intervention logic.** Treat pathogens and causal host/exposure factors; targeted microbiome therapy only where evidence supports.

**Known limits and open questions.** “Dysbiosis” is often circular and commercial tests lack actionable reference standards.

**Suite links.** 08 host-microbe; 03E/J/L/M.

*Evidence anchors:* [14,15,25]

# Assessment, clinical reasoning, and longitudinal profiling

Assessment is a staged inference process. The right test depends on the pretest question, urgency, expected change in management, burden, and the consequences of false reassurance or false alarm. A complete profile combines history, examination, validated measurements, function, trajectories, and context.

| Stage                                           | Purpose                                                                                                            | Components                                                                                                                                     | Output                                                                                       | Common failure                                                                                                     |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 1. Question and urgency                         | Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first. | Chief concern and goals; red flags; prior probability; relevant context and current biological foundations and regulation risks.               | Urgent action, focused differential, or a prespecified measurement plan.                     | Testing without a question creates incidental findings and false precision.                                        |
| 2. History and lived phenotype                  | Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.  | Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.               | A longitudinal problem representation with hypotheses and missing information.               | Administrative codes and checklists can erase chronology, context, and patient priorities.                         |
| 3. Examination and basic measurements           | Test hypotheses using standardized observation and low-burden validated measures.                                  | Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.                             | Localization, severity estimate, safety screen, and indication for further testing.          | Nonstandard technique and unblinded interpretation create systematic error.                                        |
| 4. Targeted laboratory, imaging, and physiology | Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.       | Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden. | Mechanism-informed diagnosis, prognostic information, or a revised probability distribution. | Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype. |
| 5. Function and challenge testing               | Measure reserve, adaptation, and real-world impact not visible at rest.                                            | Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.                  | Capacity, symptom coupling, recovery, and safety limits.                                     | Challenge protocols may not match daily demands and can be unsafe without screening.                               |

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| Stage                       | Purpose                                                                                                           | Components                                                                                                                     | Output                                                                   | Common failure                                                                        |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 6. Longitudinal integration | Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift. | Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds. | Trajectory, response, relapse, and need for escalation or de-escalation. | Regression to the mean and changing devices can masquerade as improvement or decline. |

### 6.0.1 1. Question and urgency

**Purpose.** Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.

**Core components.** Chief concern and goals; red flags; prior probability; relevant context and current biological foundations and regulation risks.

**Interpretation.** Urgent action, focused differential, or a prespecified measurement plan.

**Failure modes.** Testing without a question creates incidental findings and false precision.

*Evidence anchors:* <sup>[26]</sup>

### 6.0.2 2. History and lived phenotype

**Purpose.** Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.

**Core components.** Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.

**Interpretation.** A longitudinal problem representation with hypotheses and missing information.

**Failure modes.** Administrative codes and checklists can erase chronology, context, and patient priorities.

### 6.0.3 3. Examination and basic measurements

**Purpose.** Test hypotheses using standardized observation and low-burden validated measures.

**Core components.** Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.

**Interpretation.** Localization, severity estimate, safety screen, and indication for further testing.

**Failure modes.** Nonstandard technique and unblinded interpretation create systematic error.

*Evidence anchors:* <sup>[4]</sup>

#### 6.0.4 4. Targeted laboratory, imaging, and physiology

**Purpose.** Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.

**Core components.** Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.

**Interpretation.** Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.

**Failure modes.** Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.

*Evidence anchors:* <sup>[26]</sup>

#### 6.0.5 5. Function and challenge testing

**Purpose.** Measure reserve, adaptation, and real-world impact not visible at rest.

**Core components.** Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.

**Interpretation.** Capacity, symptom coupling, recovery, and safety limits.

**Failure modes.** Challenge protocols may not match daily demands and can be unsafe without screening.

#### 6.0.6 6. Longitudinal integration

**Purpose.** Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.

**Core components.** Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.

**Interpretation.** Trajectory, response, relapse, and need for escalation or de-escalation.

**Failure modes.** Regression to the mean and changing devices can masquerade as improvement or decline.

*Evidence anchors:* <sup>[24]</sup>

# Measurement and device landscape

A device does not measure health directly. It senses a physical or chemical proxy, transforms it, samples it, rejects or retains artifact, and maps the result to a claim. The claim may concern detection, diagnosis, severity, risk, response, or function; each requires a different validation design.

## Measurement is an inference chain



Validation must test every link, including population, setting, motion, missingness, drift, and downstream use.

**Figure 7.1:** Generic inference chain from latent biology to a clinical or research claim.

| Modality                        | Invasiveness                       | Signal / sensor                                                  | Outputs                                                      | Strengths                        | Limitations                                                                                   |
|---------------------------------|------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------|-----------------------------------------------------------------------------------------------|
| Karyotyping and cytogenetics    | Minimally invasive sample          | Microscopic chromosomes and FISH probes.                         | Aneuploidy, structural rearrangement and selected mosaicism. | Genome-wide structural overview. | Resolution, culture and tissue-specific mosaicism.                                            |
| DNA sequencing and genotyping   | Minimally invasive/invasive sample | Targeted, exome/genome and long-read sequence analysis.          | Variants, haplotypes and molecular diagnosis.                | High analytic resolution.        | Coverage, structural/repeat limits, variants of uncertain significance, ancestry and privacy. |
| Transcriptomics and epigenomics | Sample/research                    | RNA sequencing, expression arrays, methylation/chromatin assays. | Cell-state and regulatory patterns.                          | Dynamic biological context.      | Cell composition, timing, batch and causal ambiguity.                                         |

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| Modality                                         | Invasiveness            | Signal / sensor                                                                                         | Outputs                                                    | Strengths                                            | Limitations                                                                        |
|--------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------|
| Proteomics and functional protein assays         | Sample                  | Mass spectrometry, affinity panels, enzyme/receptor and post-translational assays.                      | Protein abundance, modification and activity.              | Closer to effector function than DNA alone.          | Dynamic range, batch, isoforms and weak reference intervals.                       |
| Metabolomics and flux analysis                   | Sample/research         | Mass spectrometry/NMR and isotope tracing.                                                              | Metabolite state and pathway flux.                         | Sensitive to active physiology.                      | Diet/time/handling, kidney/liver function and high dimensionality.                 |
| Single-cell and spatial omics                    | Sample/tissue research  | Single-cell RNA/ATAC/protein and spatial transcript/protein methods.                                    | Cell types/states, neighborhoods and trajectories.         | Resolves tissue heterogeneity.                       | Dissociation/sampling, computational choices, cost and uncertain clinical meaning. |
| Histology and molecular pathology                | Invasive tissue         | Microscopy, stains, immunohistochemistry and in situ/molecular assays.                                  | Architecture, cell injury, inflammation and neoplasia.     | Direct spatial tissue context.                       | Sampling, fixation, observer and treatment timing.                                 |
| Cellular functional assays and electrophysiology | Ex vivo/in vivo         | Stimulation, flux, microscopy, patch clamp or impedance.                                                | Receptor/channel, metabolism, secretion and cell behavior. | Mechanistic function beyond abundance.               | Artificial conditions and low throughput/standardization.                          |
| Organoids and induced pluripotent-cell models    | Ex vivo research        | Patient-derived 3D or differentiated cellular systems.                                                  | Development, disease mechanism and drug response.          | Human genetic context and experimental perturbation. | Immaturity, missing vasculature/immune/system context and batch variability.       |
| Animal and comparative models                    | Research                | Whole-organism perturbation across species.                                                             | Causal mechanism, toxicology and system interaction.       | Controlled invasive and longitudinal experiments.    | Species translation, welfare and oversimplification.                               |
| Molecular imaging and tracers                    | Noninvasive/noninvasive | PET/SPECT/MR/optical tracers targeting metabolism, receptors or cells.                                  | Spatial pathway activity and burden.                       | Whole-body in vivo biology.                          | Radiation, tracer specificity, partial volume and surrogate claims.                |
| Perturbation and challenge experiments           | Research/clinical       | Genetic, pharmacologic, nutritional, exercise or physiologic intervention with time-series measurement. | Causal response, reserve and pathway sensitivity.          | Stronger mechanistic inference than snapshots.       | Safety, compensatory networks and extrapolation.                                   |

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| Modality                                   | Invasiveness        | Signal / sensor                                                                             | Outputs                                       | Strengths              | Limitations                                                                |
|--------------------------------------------|---------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------|------------------------|----------------------------------------------------------------------------|
| Multi-omics integration and systems models | Software/researcher | Linked molecular, cellular, clinical and exposure data with statistical/mechanistic models. | Candidate pathways, subtypes and predictions. | Cross-scale synthesis. | Overfitting, batch/confounding, unidentifiable models and opaque clusters. |

### 7.0.1 Karyotyping and cytogenetics

**Measured quantity and transducer.** Microscopic chromosomes and FISH probes.

**Typical outputs.** Aneuploidy, structural rearrangement and selected mosaicism.

**Strengths.** Genome-wide structural overview.

**Limitations and artifacts.** Resolution, culture and tissue-specific mosaicism.

**Reference and validation strategy.** Controls, orthogonal confirmation and phenotype correlation.

**Clinical/research maturity.** Established.

*Evidence anchors:* [8,27]

### 7.0.2 DNA sequencing and genotyping

**Measured quantity and transducer.** Targeted, exome/genome and long-read sequence analysis.

**Typical outputs.** Variants, haplotypes and molecular diagnosis.

**Strengths.** High analytic resolution.

**Limitations and artifacts.** Coverage, structural/repeat limits, variants of uncertain significance, ancestry and privacy.

**Reference and validation strategy.** Analytic validation, confirmatory methods and clinical utility.

**Clinical/research maturity.** Established selected; expanding.

*Evidence anchors:* [8,24]

### 7.0.3 Transcriptomics and epigenomics

**Measured quantity and transducer.** RNA sequencing, expression arrays, methylation/chromatin assays.

**Typical outputs.** Cell-state and regulatory patterns.

**Strengths.** Dynamic biological context.

**Limitations and artifacts.** Cell composition, timing, batch and causal ambiguity.

**Reference and validation strategy.** Standard preanalytics, external replication and perturbation/utility.

**Clinical/research maturity.** Research; selected clinical signatures.

*Evidence anchors:* [6,7]

## 7.0.4 Proteomics and functional protein assays

**Measured quantity and transducer.** Mass spectrometry, affinity panels, enzyme/receptor and post-translational assays.

**Typical outputs.** Protein abundance, modification and activity.

**Strengths.** Closer to effector function than DNA alone.

**Limitations and artifacts.** Dynamic range, batch, isoforms and weak reference intervals.

**Reference and validation strategy.** Reference materials, orthogonal function and clinical utility.

**Clinical/research maturity.** Established narrow; research broad.

*Evidence anchors:* [7,16]

## 7.0.5 Metabolomics and flux analysis

**Measured quantity and transducer.** Mass spectrometry/NMR and isotope tracing.

**Typical outputs.** Metabolite state and pathway flux.

**Strengths.** Sensitive to active physiology.

**Limitations and artifacts.** Diet/time/handling, kidney/liver function and high dimensionality.

**Reference and validation strategy.** Controlled preanalytics, repeated measures and intervention causality.

**Clinical/research maturity.** Research; selected newborn/metabolic diagnostics.

*Evidence anchors:* [7,24]

### 7.0.6 Single-cell and spatial omics

**Measured quantity and transducer.** Single-cell RNA/ATAC/protein and spatial transcript/protein methods.

**Typical outputs.** Cell types/states, neighborhoods and trajectories.

**Strengths.** Resolves tissue heterogeneity.

**Limitations and artifacts.** Dissociation/sampling, computational choices, cost and uncertain clinical meaning.

**Reference and validation strategy.** Cross-platform, histology/function and prospective utility.

**Clinical/research maturity.** Advanced research.

*Evidence anchors:* <sup>[6,7]</sup>

### 7.0.7 Histology and molecular pathology

**Measured quantity and transducer.** Microscopy, stains, immunohistochemistry and in situ/molecular assays.

**Typical outputs.** Architecture, cell injury, inflammation and neoplasia.

**Strengths.** Direct spatial tissue context.

**Limitations and artifacts.** Sampling, fixation, observer and treatment timing.

**Reference and validation strategy.** Expert integrated reference and outcome correlation.

**Clinical/research maturity.** Established.

*Evidence anchors:* <sup>[4,20]</sup>

### 7.0.8 Cellular functional assays and electrophysiology

**Measured quantity and transducer.** Stimulation, flux, microscopy, patch clamp or impedance.

**Typical outputs.** Receptor/channel, metabolism, secretion and cell behavior.

**Strengths.** Mechanistic function beyond abundance.

**Limitations and artifacts.** Artificial conditions and low throughput/standardization.

**Reference and validation strategy.** Reference cells/materials and genotype/phenotype/outcome links.

**Clinical/research maturity.** Established selected/research.

*Evidence anchors:* [3,16]

### 7.0.9 Organoids and induced pluripotent-cell models

**Measured quantity and transducer.** Patient-derived 3D or differentiated cellular systems.

**Typical outputs.** Development, disease mechanism and drug response.

**Strengths.** Human genetic context and experimental perturbation.

**Limitations and artifacts.** Immaturity, missing vasculature/immune/system context and batch variability.

**Reference and validation strategy.** Benchmark to primary tissue/clinical response and reproducibility.

**Clinical/research maturity.** Research/emerging.

*Evidence anchors:* [6,8]

### 7.0.10 Animal and comparative models

**Measured quantity and transducer.** Whole-organism perturbation across species.

**Typical outputs.** Causal mechanism, toxicology and system interaction.

**Strengths.** Controlled invasive and longitudinal experiments.

**Limitations and artifacts.** Species translation, welfare and oversimplification.

**Reference and validation strategy.** Model validity, triangulation with human evidence and preregistration.

**Clinical/research maturity.** Established research.

*Evidence anchors:* [23,30]

### 7.0.11 Molecular imaging and tracers

**Measured quantity and transducer.** PET/SPECT/MR/optical tracers targeting metabolism, receptors or cells.

**Typical outputs.** Spatial pathway activity and burden.

**Strengths.** Whole-body in vivo biology.

**Limitations and artifacts.** Radiation, tracer specificity, partial volume and surrogate claims.

**Reference and validation strategy.** Chemistry, blocking/competition, pathology and outcome validation.

**Clinical/research maturity.** Established selected/research.

*Evidence anchors:* [20,28]

### 7.0.12 Perturbation and challenge experiments

**Measured quantity and transducer.** Genetic, pharmacologic, nutritional, exercise or physiologic intervention with time-series measurement.

**Typical outputs.** Causal response, reserve and pathway sensitivity.

**Strengths.** Stronger mechanistic inference than snapshots.

**Limitations and artifacts.** Safety, compensatory networks and extrapolation.

**Reference and validation strategy.** Controls, dose/time, preregistration and replication.

**Clinical/research maturity.** Established research principle.

*Evidence anchors:* [22,23]

### 7.0.13 Multi-omics integration and systems models

**Measured quantity and transducer.** Linked molecular, cellular, clinical and exposure data with statistical/mechanistic models.

**Typical outputs.** Candidate pathways, subtypes and predictions.

**Strengths.** Cross-scale synthesis.

**Limitations and artifacts.** Overfitting, batch/confounding, unidentifiable models and opaque clusters.

**Reference and validation strategy.** Independent cohorts, perturbation, calibration and decision utility.

**Clinical/research maturity.** Frontier.

*Evidence anchors:* [7,24,25]

# Engineering architectures, signal quality, and validation

Buildable systems should begin with an explicit intended use, measurand, reference method, population, environment, and failure policy. Prototype plausibility is not clinical validity. Raw data, quality channels, timing, calibration state, firmware version, and provenance should be retained so that errors can be reconstructed rather than hidden by a polished score.

| Subsystem / design issue                   | Requirements                                                                                                                                                         | Dominant risks                                                                                                                    | Verification                                                                                                                |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Intended use and measurand                 | Define the precise biological foundations and regulation quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.               | A broad wellness claim can hide several incompatible clinical claims.                                                             | Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.              |
| Sensor and transduction                    | Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.            | A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing. | Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range. |
| Timing and synchronization                 | Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.                               | Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.                                      | Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.                   |
| Analog, acquisition, and digital integrity | Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.                              | Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.                              | Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.                |
| Calibration and traceability               | Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.                                       | Model calibration to one device lot, site, or operator may not transport.                                                         | Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.           |
| Artifact and quality channels              | Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal. | Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.                        | Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.                 |

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| Subsystem / design issue                      | Requirements                                                                                                                                               | Dominant risks                                                                                               | Verification                                                                                                                             |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Algorithms and uncertainty                    | Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.                                    | Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.           | Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing. |
| Safety, usability, cybersecurity, and privacy | Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization. | Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays. | Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring. |

### 8.0.1 Intended use and measurand

**Design requirements.** Define the precise biological foundations and regulation quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.

**Failure modes.** A broad wellness claim can hide several incompatible clinical claims.

**Bench verification.** Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.

**Human validation boundary.** Do not enroll people until the measurement purpose and risk controls are reviewable.

### 8.0.2 Sensor and transduction

**Design requirements.** Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.

**Failure modes.** A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.

**Bench verification.** Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.

**Human validation boundary.** Bench testing should precede contact with participants.

### 8.0.3 Timing and synchronization

**Design requirements.** Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.

**Failure modes.** Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.

**Bench verification.** Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.

**Human validation boundary.** Store both device time and host/reference time.

*Evidence anchors:* <sup>[24]</sup>

#### 8.0.4 Analog, acquisition, and digital integrity

**Design requirements.** Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.

**Failure modes.** Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.

**Bench verification.** Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.

**Human validation boundary.** Preserve raw or minimally processed channels where ethically and technically possible.

*Evidence anchors:* <sup>[27]</sup>

#### 8.0.5 Calibration and traceability

**Design requirements.** Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.

**Failure modes.** Model calibration to one device lot, site, or operator may not transport.

**Bench verification.** Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.

**Human validation boundary.** Calibration data and version must travel with each observation.

*Evidence anchors:* <sup>[24,27]</sup>

#### 8.0.6 Artifact and quality channels

**Design requirements.** Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.

**Failure modes.** Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.

**Bench verification.** Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

**Human validation boundary.** Quality rejection must not silently exclude populations.

*Evidence anchors:* <sup>[25]</sup>

### 8.0.7 Algorithms and uncertainty

**Design requirements.** Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.

**Failure modes.** Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.

**Bench verification.** Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.

**Human validation boundary.** A model should abstain when inputs are outside validated conditions.

*Evidence anchors:* <sup>[25]</sup>

### 8.0.8 Safety, usability, cybersecurity, and privacy

**Design requirements.** Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.

**Failure modes.** Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.

**Bench verification.** Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

**Human validation boundary.** Research exemptions do not remove the ethical duty to minimize foreseeable harm.

*Evidence anchors:* <sup>[30]</sup>

#### Human-contact engineering rule

Do not connect a human participant to a mains-referenced oscilloscope, programmer, charger, or non-medical ground path. Bench characterization should use simulators and isolated fixtures. Human research requires documented electrical safety, risk management, consent, privacy controls, and appropriate oversight.

# Prevention, treatment, rehabilitation, and system intervention

Interventions are classified by target and level: upstream determinants, exposure reduction, behavioral support, prevention, diagnosis-linked treatment, acute rescue, chronic management, rehabilitation, assistive technology, and palliative or supportive care. Net benefit depends on baseline risk, absolute effect, burden, interactions, access, adherence, and the outcomes valued by the person.

| Intervention                          | Level                | Target / mechanism                                                                             | Evidence profile                                                                          | Risks / burdens                                                     | Monitoring                                                                    |
|---------------------------------------|----------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Exposure and developmental prevention | Upstream prevention  | Reduce known mutagens, teratogens, toxins, infection and deprivation during sensitive periods. | Strong for selected exposures; often larger benefit than downstream molecular correction. | Unequal feasibility and substitution effects.                       | Exposure reduction, developmental/organ outcomes and equity.                  |
| Nutrient and cofactor replacement     | Replacement          | Restore a deficient substrate required for enzyme, blood, neural or tissue function.           | Transformative when a true deficiency or metabolic block is identified.                   | Overdose, missed cause and nonspecific marketing.                   | Biochemical/function response and toxicity.                                   |
| Small-molecule pathway modulation     | Treatment            | Inhibit, activate or stabilize enzymes, channels, receptors or protein interactions.           | Core pharmacology with broad evidence across diseases.                                    | Off-target effects, interactions, resistance and variable exposure. | PK/PD, target/clinical response and safety.                                   |
| Biologic and antibody therapy         | Treatment/prevention | Replace proteins or target extracellular/cellular pathways with large molecules.               | High specificity and major benefits in selected disease.                                  | Immune reactions, infection, delivery, cost and escape.             | Exposure, immune safety, target/clinical outcomes and long-term surveillance. |
| Hormone and physiologic replacement   | Replacement          | Restore missing endocrine or organ product/function.                                           | Effective for clearly defined deficiency.                                                 | Overreplacement, timing and organ interactions.                     | Symptoms, target hormones/function and adverse effects.                       |
| Cellular therapy and transplantation  | Advanced/cure        | Replace cells/tissue or provide immune/functional capacity.                                    | Curative or transformative in selected hematologic and organ states.                      | Rejection, graft disease, infection, malignancy and access.         | Engraftment/function, immune risk and long-term outcomes.                     |

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| Intervention                                   | Level               | Target / mechanism                                                                          | Evidence profile                                                                                                   | Risks / burdens                                                      | Monitoring                                                            |
|------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------|
| Gene addition, editing and RNA therapy         | Advanced/precision  | Alter gene expression or sequence in vivo/ex vivo.                                          | Established for a growing but narrow set of diseases.                                                              | Off-target, immune, delivery, clonal, germline and durability risks. | Molecular correction, function and long-term surveillance.            |
| Proteostasis and degradation therapy           | Treatment/frontier  | Stabilize proteins, enhance folding/clearance or recruit targeted degradation.              | Effective for selected mutations/cancers; broad neurodegenerative benefit remains limited.                         | Off-target degradation and network effects.                          | Target engagement, function and toxicity.                             |
| Regenerative and tissue-engineering strategies | Advanced/frontier   | Use cells, scaffolds, factors or bioprinting to restore tissue.                             | Established for narrow tissues/products; many claims remain experimental.                                          | Immune, tumor, ectopic growth, manufacturing and mechanical failure. | Product identity/potency, integration, function and long-term safety. |
| Microbiome-directed intervention               | Treatment/precision | Use pathogen eradication, diet, defined organisms/products or ecosystem transfer.           | Established for selected infections; evidence is mixed/early for many chronic diseases.                            | Infection, ecological effects, AMR, product variability and hype.    | Microbial and clinical outcomes with long-term safety.                |
| Senescence/aging targeted intervention         | Frontier            | Modify damage, nutrient sensing, senescence, autophagy or regenerative decline.             | Strong model evidence; human clinical benefit is indication-specific and mostly unproven for general “anti-aging.” | Cancer, immune, metabolic and unknown long-term harms.               | Patient-important outcomes, not only molecular-age markers.           |
| Mechanism-informed combination therapy         | Systems treatment   | Target multiple nodes or pair causal treatment with supportive/rehabilitative intervention. | Often necessary in cancer, infection and complex disease.                                                          | Interactions, toxicity, attribution and combinatorial evidence gaps. | Component contribution, exposure, outcomes and adaptive resistance.   |

### 9.0.1 Exposure and developmental prevention

**Target and mechanism.** Reduce known mutagens, teratogens, toxins, infection and deprivation during sensitive periods.

**Evidence and expected outcome.** Strong for selected exposures; often larger benefit than downstream molecular correction.

**Risks, burden, and equity.** Unequal feasibility and substitution effects.

**Response monitoring.** Exposure reduction, developmental/organ outcomes and equity.

*Evidence anchors:* [12,21]

### 9.0.2 Nutrient and cofactor replacement

**Target and mechanism.** Restore a deficient substrate required for enzyme, blood, neural or tissue function.

**Evidence and expected outcome.** Transformative when a true deficiency or metabolic block is identified.

**Risks, burden, and equity.** Overdose, missed cause and nonspecific marketing.

**Response monitoring.** Biochemical/function response and toxicity.

*Evidence anchors:* [5,26]

### 9.0.3 Small-molecule pathway modulation

**Target and mechanism.** Inhibit, activate or stabilize enzymes, channels, receptors or protein interactions.

**Evidence and expected outcome.** Core pharmacology with broad evidence across diseases.

**Risks, burden, and equity.** Off-target effects, interactions, resistance and variable exposure.

**Response monitoring.** PK/PD, target/clinical response and safety.

*Evidence anchors:* [16,28,29]

### 9.0.4 Biologic and antibody therapy

**Target and mechanism.** Replace proteins or target extracellular/cellular pathways with large molecules.

**Evidence and expected outcome.** High specificity and major benefits in selected disease.

**Risks, burden, and equity.** Immune reactions, infection, delivery, cost and escape.

**Response monitoring.** Exposure, immune safety, target/clinical outcomes and long-term surveillance.

*Evidence anchors:* [8,28]

### 9.0.5 Hormone and physiologic replacement

**Target and mechanism.** Restore missing endocrine or organ product/function.

**Evidence and expected outcome.** Effective for clearly defined deficiency.

**Risks, burden, and equity.** Overreplacement, timing and organ interactions.

**Response monitoring.** Symptoms, target hormones/function and adverse effects.

*Evidence anchors:* [5,17]

### 9.0.6 Cellular therapy and transplantation

**Target and mechanism.** Replace cells/tissue or provide immune/functional capacity.

**Evidence and expected outcome.** Curative or transformative in selected hematologic and organ states.

**Risks, burden, and equity.** Rejection, graft disease, infection, malignancy and access.

**Response monitoring.** Engraftment/function, immune risk and long-term outcomes.

*Evidence anchors:* [8,20]

### 9.0.7 Gene addition, editing and RNA therapy

**Target and mechanism.** Alter gene expression or sequence in vivo/ex vivo.

**Evidence and expected outcome.** Established for a growing but narrow set of diseases.

**Risks, burden, and equity.** Off-target, immune, delivery, clonal, germline and durability risks.

**Response monitoring.** Molecular correction, function and long-term surveillance.

*Evidence anchors:* [8,30]

### 9.0.8 Proteostasis and degradation therapy

**Target and mechanism.** Stabilize proteins, enhance folding/clearance or recruit targeted degradation.

**Evidence and expected outcome.** Effective for selected mutations/cancers; broad neurodegenerative benefit remains limited.

**Risks, burden, and equity.** Off-target degradation and network effects.

**Response monitoring.** Target engagement, function and toxicity.

*Evidence anchors:* [9,16]

### 9.0.9 Regenerative and tissue-engineering strategies

**Target and mechanism.** Use cells, scaffolds, factors or bioprinting to restore tissue.

**Evidence and expected outcome.** Established for narrow tissues/products; many claims remain experimental.

**Risks, burden, and equity.** Immune, tumor, ectopic growth, manufacturing and mechanical failure.

**Response monitoring.** Product identity/potency, integration, function and long-term safety.

*Evidence anchors:* [27,30]

### 9.0.10 Microbiome-directed intervention

**Target and mechanism.** Use pathogen eradication, diet, defined organisms/products or ecosystem transfer.

**Evidence and expected outcome.** Established for selected infections; evidence is mixed/early for many chronic diseases.

**Risks, burden, and equity.** Infection, ecological effects, AMR, product variability and hype.

**Response monitoring.** Microbial and clinical outcomes with long-term safety.

*Evidence anchors:* [14,15]

### 9.0.11 Senescence/aging-targeted intervention

**Target and mechanism.** Modify damage, nutrient sensing, senescence, autophagy or regenerative decline.

**Evidence and expected outcome.** Strong model evidence; human clinical benefit is indication-specific and mostly unproven for general “anti-aging.”

**Risks, burden, and equity.** Cancer, immune, metabolic and unknown long-term harms.

**Response monitoring.** Patient-important outcomes, not only molecular-age markers.

*Evidence anchors:* [9,26]

### 9.0.12 Mechanism-informed combination therapy

**Target and mechanism.** Target multiple nodes or pair causal treatment with supportive/rehabilitative intervention.

**Evidence and expected outcome.** Often necessary in cancer, infection and complex disease.

**Risks, burden, and equity.** Interactions, toxicity, attribution and combinatorial evidence gaps.

**Response monitoring.** Component contribution, exposure, outcomes and adaptive resistance.

*Evidence anchors:* [11,29]

# What is known, not known, and frequently overstated

Uncertainty is not a single residual. It can arise from biological variability, measurement error, selection, confounding, limited follow-up, inadequate reference standards, endpoint substitution, poor external validity, causal ambiguity, or value disagreement. The correct response is to identify the uncertainty and design the next measurement or study.

| Question                                       | What is established                                                                                                   | What remains uncertain                                                                                                                           | Why unresolved                                                                                                             | Decisive next step                                                                                                                      |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Can biological age be measured meaningfully?   | Molecular clocks correlate with chronological age and selected outcomes.                                              | Whether changing a clock changes health and which clock represents causal aging.                                                                 | Training outcomes, tissue mixture and reverse causality.                                                                   | Intervention trials with patient-important outcomes and mechanistic mediation.                                                          |
| Can multi-omics define actionable subtypes?    | High-dimensional data reveal reproducible patterns in some diseases.                                                  | Stable treatment-interaction benefit beyond clinical variables.                                                                                  | Batch, cohort and algorithm choices create unstable clusters.                                                              | Locked assays and prospective subtype-by-treatment trials.                                                                              |
| How much risk is genetic versus environmental? | Both inherited variation and environments contribute, often through interaction and mediation across the life course. | No single percentage applies to all outcomes, populations, ages, or definitions within biological foundations and regulation.                    | Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap. | Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands. |
| Can one composite score summarize health?      | Composite scores can rank a defined outcome in a defined population and may improve workflow.                         | Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions. | Different variables compensate, interact, and have different treatment implications; weights drift over time.              | Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.       |

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| Question                              | What is established                                                                                                          | What remains uncertain                                                                                                  | Why unresolved                                                                                    | Decisive next step                                                                                                   |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| When is earlier detection beneficial? | Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms. | Net benefit for many incidental, molecular, wearable, and multi-cancer signals.                                         | Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.     | Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis. |
| Do digital biomarkers generalize?     | Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.                             | Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity. | Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels. | Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.  |
| What counts as meaningful change?     | Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.                 | Thresholds for many biomarkers, sensors, and multidomain profiles.                                                      | Baseline severity, goals, adaptation, and treatment burden alter value.                           | Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.          |

### 10.0.1 Can biological age be measured meaningfully?

**Established knowledge.** Molecular clocks correlate with chronological age and selected outcomes.

**Unresolved issue.** Whether changing a clock changes health and which clock represents causal aging.

**Why current evidence is insufficient.** Training outcomes, tissue mixture and reverse causality.

**Research that would change confidence.** Intervention trials with patient-important outcomes and mechanistic mediation.

*Evidence anchors:* [9,25]

### 10.0.2 Can multi-omics define actionable subtypes?

**Established knowledge.** High-dimensional data reveal reproducible patterns in some diseases.

**Unresolved issue.** Stable treatment-interaction benefit beyond clinical variables.

**Why current evidence is insufficient.** Batch, cohort and algorithm choices create unstable clusters.

**Research that would change confidence.** Locked assays and prospective subtype-by-treatment trials.

*Evidence anchors:* [7,8]

### 10.0.3 How much risk is genetic versus environmental?

**Established knowledge.** Both inherited variation and environments contribute, often through interaction and mediation across the life course.

**Unresolved issue.** No single percentage applies to all outcomes, populations, ages, or definitions within biological foundations and regulation.

**Why current evidence is insufficient.** Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.

**Research that would change confidence.** Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.

*Evidence anchors:* [8,22]

### 10.0.4 Can one composite score summarize health?

**Established knowledge.** Composite scores can rank a defined outcome in a defined population and may improve workflow.

**Unresolved issue.** Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.

**Why current evidence is insufficient.** Different variables compensate, interact, and have different treatment implications; weights drift over time.

**Research that would change confidence.** Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

*Evidence anchors:* [25]

### 10.0.5 When is earlier detection beneficial?

**Established knowledge.** Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.

**Unresolved issue.** Net benefit for many incidental, molecular, wearable, and multi-cancer signals.

**Why current evidence is insufficient.** Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.

**Research that would change confidence.** Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.

*Evidence anchors:* [20,26]

### 10.0.6 Do digital biomarkers generalize?

**Established knowledge.** Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.

**Unresolved issue.** Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.

**Why current evidence is insufficient.** Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.

**Research that would change confidence.** Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.

### 10.0.7 What counts as meaningful change?

**Established knowledge.** Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.

**Unresolved issue.** Thresholds for many biomarkers, sensors, and multidomain profiles.

**Why current evidence is insufficient.** Baseline severity, goals, adaptation, and treatment burden alter value.

**Research that would change confidence.** Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

# Research and development roadmap

The roadmap prioritizes work that changes decisions or reduces uncertainty, not merely work that generates more variables. Each project should declare the causal or measurement target, comparator, reference, population, sampling frame, missing-data plan, subgroup analysis, and stopping criteria before data collection.

| Priority                          | Question                                                                                                                                                                           | Minimum credible design                                                                                                                                             | Success criterion                                                                            | Stop / failure criterion                                                            |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 0. Cross-scale perturbation atlas | Which molecular/cellular responses predict organ reserve and recovery?                                                                                                             | Standardized human challenge studies with multi-omics, physiology and function over time.                                                                           | Replicated causal mediators and improved intervention targeting.                             | Static associations or unidentifiable models.                                       |
| 0. Translation failure registry   | Why do strong preclinical mechanisms fail in humans?                                                                                                                               | Prospective preregistered tracking of target validity, exposure, model assumptions and trial outcomes.                                                              | Identifies modifiable failure patterns and improves portfolio decisions.                     | Retrospective narrative without denominator.                                        |
| 1. Reference map                  | What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for biological foundations and regulation? | Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups. | Reproducible distributions and state-dependent reference models with quantified uncertainty. | Convenience sample presented as universal normality.                                |
| 2. Causal determinants            | Which modifiable exposures cause clinically and functionally important outcomes, and in whom?                                                                                      | Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.   | Convergent effect estimates linked to plausible mechanisms and actionable interventions.     | Association-only result marketed as prevention guidance.                            |
| 3. Measurement validity           | Does the new device or biomarker measure the intended quantity under real use?                                                                                                     | Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.                       | Prespecified error and failure limits met across intended-use conditions.                    | High average correlation with systematic individual error or unreported exclusions. |
| 4. Clinical utility               | Does using the measurement improve decisions or outcomes compared with current care?                                                                                               | Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.                                                  | Net benefit on patient-important outcomes with acceptable implementation burden.             | Improved prediction without changed decisions or outcomes.                          |

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| Priority                         | Question                                                                                     | Minimum credible design                                                                                                                                   | Success criterion                                                               | Stop / failure criterion                                                    |
|----------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 5. Longitudinal multimorbidity   | How does the domain interact with other conditions, treatments, and functional trajectories? | Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks. | Models that support coordinated care rather than isolated disease optimization. | Single-disease endpoint worsens overall burden or conflicts with goals.     |
| 6. Implementation and governance | Can effective interventions be delivered safely, equitably, affordably, and sustainably?     | Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.       | Sustained adoption with monitored outcomes and reduced inequity.                | Pilot success depends on exceptional staffing or hidden externalized costs. |

### 11.0.1 0. Cross-scale perturbation atlas

**Question.** Which molecular/cellular responses predict organ reserve and recovery?

**Design.** Standardized human challenge studies with multi-omics, physiology and function over time.

**Success criterion.** Replicated causal mediators and improved intervention targeting.

**Failure or stop criterion.** Static associations or unidentifiable models.

*Evidence anchors:* [6,24]

### 11.0.2 0. Translation failure registry

**Question.** Why do strong preclinical mechanisms fail in humans?

**Design.** Prospective preregistered tracking of target validity, exposure, model assumptions and trial outcomes.

**Success criterion.** Identifies modifiable failure patterns and improves portfolio decisions.

**Failure or stop criterion.** Retrospective narrative without denominator.

*Evidence anchors:* [23,26]

### 11.0.3 1. Reference map

**Question.** What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for biological foundations and regulation?

**Design.** Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.

**Success criterion.** Reproducible distributions and state-dependent reference models with quantified uncertainty.

**Failure or stop criterion.** Convenience sample presented as universal normality.

*Evidence anchors:* [24]

## 11.0.4 2. Causal determinants

**Question.** Which modifiable exposures cause clinically and functionally important outcomes, and in whom?

**Design.** Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.

**Success criterion.** Convergent effect estimates linked to plausible mechanisms and actionable interventions.

**Failure or stop criterion.** Association-only result marketed as prevention guidance.

*Evidence anchors:* [22,23]

## 11.0.5 3. Measurement validity

**Question.** Does the new device or biomarker measure the intended quantity under real use?

**Design.** Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.

**Success criterion.** Prespecified error and failure limits met across intended-use conditions.

**Failure or stop criterion.** High average correlation with systematic individual error or unreported exclusions.

*Evidence anchors:* [25]

## 11.0.6 4. Clinical utility

**Question.** Does using the measurement improve decisions or outcomes compared with current care?

**Design.** Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.

**Success criterion.** Net benefit on patient-important outcomes with acceptable implementation burden.

**Failure or stop criterion.** Improved prediction without changed decisions or outcomes.

*Evidence anchors:* [26]

### 11.0.7 5. Longitudinal multimorbidity

**Question.** How does the domain interact with other conditions, treatments, and functional trajectories?

**Design.** Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.

**Success criterion.** Models that support coordinated care rather than isolated disease optimization.

**Failure or stop criterion.** Single-disease endpoint worsens overall burden or conflicts with goals.

### 11.0.8 6. Implementation and governance

**Question.** Can effective interventions be delivered safely, equitably, affordably, and sustainably?

**Design.** Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.

**Success criterion.** Sustained adoption with monitored outcomes and reduced inequity.

**Failure or stop criterion.** Pilot success depends on exceptional staffing or hidden externalized costs.

# Cross-volume integration

The health atlas is deliberately polyhierarchical. A phenomenon can have one primary home while also participating in multiple causal and functional systems. Cross-references below identify the most important companion volumes and the reason for the link.

| Volume | Companion topic                                   | Relationship                                                                                                 | Use this link when...                                                                     |
|--------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 00     | Atlas of Human Health                             | Master taxonomy, history of classification, and multi-axial architecture.                                    | locating the primary home of a concept or mapping classifications.                        |
| 13     | Measurement, Evidence, Data, and Informatics      | Shared metrology, causal inference, validation, interoperability, and data governance.                       | designing a study, sensor, dataset, model, or evidence claim.                             |
| 14     | Interventions and Therapeutics                    | Cross-domain pharmacology, procedures, behavioral interventions, rehabilitation, implementation, and safety. | moving from measurement to action.                                                        |
| 15     | Ethics, Law, Rights, Equity, and Governance       | Consent, privacy, fairness, rights, research oversight, and distribution of benefits and harms.              | people, identifiable data, invasive procedures, or consequential algorithms are involved. |
| 030    | Integrated Homeostatic Systems                    | Translates molecular/cellular processes into regulated whole-organism dynamics.                              | mechanism must be connected to physiology and challenge.                                  |
| 04     | Cross-System Pathological Processes               | Inflammation, fibrosis, thrombosis, cancer, degeneration and injury are emergent failure programs.           | a process spans several organs.                                                           |
| 07     | Development and Life Course                       | Sensitive periods, plasticity, aging and cumulative exposure change biological meaning.                      | timing is causal.                                                                         |
| 08     | Infection, Immunity, Host-Microbe and Biosecurity | Microbes and host defense are ecological and evolutionary biological systems.                                | host-microbe mechanisms are central.                                                      |
| 13     | Measurement, Evidence, Data and Informatics       | Provides metrology, causal inference and validation for high-dimensional biology.                            | turning omics into a claim or decision.                                                   |

## Cross-reference rule

Reference the primary home for the full mechanism or standard; restate only what is necessary for the local argument. When a claim spans domains, record the bridge explicitly rather than assigning the entire phenomenon to one specialty.

# Minimum data schema for a longitudinal domain profile

| Layer                     | Minimum fields                                                                                                                       | Critical caution                                                                      |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Identity and context      | Participant identifier, consent, demographics relevant to the question, setting, language, access, and social/environmental context. | Avoid collecting identity variables without purpose; preserve privacy and provenance. |
| Time                      | Absolute timestamps, timezone, age/life stage, episode boundaries, event markers, medication and exposure timing.                    | Clock drift and ambiguous time zones can invalidate multimodal inference.             |
| Raw signal / specimen     | Unprocessed samples, units, sensor channel, specimen type, collection protocol, calibration and storage state.                       | Derived scores without raw or quality data are difficult to audit.                    |
| Quality                   | Contact, motion, saturation, missingness, assay controls, operator notes, confidence, out-of-range flags.                            | Quality must be a first-class variable, not only an exclusion after analysis.         |
| Reference                 | Comparator method, instrument, version, observer protocol, laboratory traceability, adjudication.                                    | A weak reference caps the validity of the new system.                                 |
| Phenotype and function    | Symptoms, signs, diagnoses, severity, capability, participation, patient-reported outcomes, goals.                                   | Disease codes alone do not represent lived outcome.                                   |
| Intervention and response | Dose, route, device settings, behavior support, adherence, co-interventions, adverse events, response window.                        | Confounding by indication and treatment changes must be modeled.                      |
| Governance                | Protocol version, data-use agreement, access log, retention, deletion, model version, incident record.                               | Data stewardship is part of scientific validity.                                      |

# Glossary

| Term                | Operational meaning in this volume                                                                                                                    |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Epigenetics         | Heritable or stable regulation of gene activity not caused by change in DNA sequence; usage should specify mechanism and cell context.                |
| Proteostasis        | Networks controlling protein synthesis, folding, trafficking and degradation.                                                                         |
| Mechanotransduction | Conversion of mechanical force or geometry into cellular biochemical response.                                                                        |
| Multi-omics         | Integrated measurement of multiple molecular layers; not inherently causal or clinically useful.                                                      |
| Emergence           | System behavior arising from interactions among components and not adequately predicted by isolated parts.                                            |
| Allostasis          | Adaptive regulation that changes physiological activity to meet anticipated or actual demands; chronic or repeated activation can carry cost.         |
| Biomarker           | Defined characteristic measured as an indicator of normal biology, pathogenic process, exposure, or response; not automatically a surrogate endpoint. |
| Function            | What a person can do in body functions, activities, and participation, interpreted in context.                                                        |
| Homeostasis         | Regulation of variables within viable ranges through feedback, feedforward, redundancy, and adaptation.                                               |
| Phenotype           | Observable or measurable characteristics arising from biology, development, environment, behavior, context, and measurement conditions.               |
| Risk factor         | Characteristic associated with future outcome; causal status and modifiability must be established separately.                                        |
| Calibration         | Establishing the relationship between an instrument output and a reference quantity under specified conditions.                                       |
| Clinical validity   | Degree to which a measurement or model is associated with or predicts the clinical state or outcome it claims to represent.                           |
| Intended use        | The specific purpose, population, setting, user, input, and decision for which a device or model is designed.                                         |
| Reference method    | Comparator selected to estimate truth for validation; every reference has its own error and scope.                                                    |
| Reserve             | Capacity to maintain function under challenge beyond resting baseline.                                                                                |
| Trajectory          | Direction, rate, variability, and reversibility of change over time.                                                                                  |

# Selected references and current standards

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The bibliography emphasizes official standards, current guidelines, landmark primary work, and major syntheses. Inclusion does not imply that every recommendation is universal or that institutional endorsement removes uncertainty. Current online materials were checked through the stated evidence cutoff.

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# Version note

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Foundation edition generated August 2026. Evidence, guidance, and standards cutoff: 13 August 2026. The suite should be versioned when major classifications, guidelines, device standards, or decisive evidence materially change the map.